Gaurang Ranjan Singh

Roll No.: 12311062
Bachelor of Technology
Indian Institute of Information Technology (IIIT) Sonepat

+91-63924 42962
gaurangiiit@gmail.com
GitHub Profile
LinkedIn Profile

Education

•Bachelor of Technology in Computer Science and Engineering *Aug 2023 - Expected 2027*
Indian Institute of Information Technology (IIIT) Sonepat CGPA: 8.55/10

•High School (Class 12: 96%) | ICSE (Class 10: 98.4%) *2021 - 2023*
City Montessori School, Lucknow

Projects & Research

•Predicting Traffic Congestion Using Machine Learning *Oct 2025*
Academic Research Paper on Spatiotemporal Forecasting.

- Analyzed over 1 billion NYC Taxi records leveraging H3 geospatial indexing to uncover spatiotemporal congestion patterns.
- Architected predictive models (RF, XGBoost, LSTM), with the optimized deep learning network achieving an industry-leading R^2 score of 0.91 and RMSE of 0.129.
- Simulated predictions within the SUMO traffic environment, demonstrating an 18.3% enhancement in average travel time and a 12.1% reduction in intersection delays.

•Dynamic-SLAM *GitHub*
Dynamic Simultaneous Localization and Mapping implementation.

- Constructed a real-time SLAM pipeline using Python, processing 30+ frames per second for accurate environment rendering.
- Engineered algorithmic tracking modules and data visualization libraries, securing up to 95% mapping accuracy during complex state estimations.

•Manorath Event Manager *GitHub*
Comprehensive event management application.

- Built a scalable event scheduling application using Dart and Flutter, supporting over 100 concurrent user schedules.
- Accelerated event coordination by implementing interactive UI components and state management, reducing average booking time by 40%.

•Gesture-Specialist *GitHub*
Interactive web utility for hand gesture recognition.

- Designed an interactive web utility with HTML, CSS, and JavaScript to precisely recognize and process 10+ specialized hand gestures.
- Maximized application performance by optimizing event listeners and DOM manipulation, driving system response latency under 50ms.

Leadership & Teamwork

•Coding Mentor: Instructed 5+ junior students in C++ and Data Structures, conducting technical workshops to elevate algorithm optimization and competitive programming skills.

•Institute Hackathon: Programmed a real-time chat feature alongside 2 peers during an intensive 24-hour event, delivering the working prototype to an audience of 50+ attendees.

Technical Skills

Languages: Java, C/C++, Python, JavaScript, Dart, SQL, HTML/CSS
Libraries & Frameworks: TensorFlow/Keras, React.js, Node.js, Tailwind, NumPy, Pandas, Scikit-learn
Tools & Databases: Git, GitHub, VS Code, Postman, MySQL, SUMO, LaTeX
Relevant Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Theory of Computation, Compiler Design, Machine Learning

Certifications & Achievements

•GATE 2026 Qualified: Computer Science and Information Technology (CSE/IT) | Data Science and AI (DA/AI)

•Competitive Programming: Codeforces Pupil (Max Rating: 1170, 300+ Solved) | CodeChef 2-Star *2023 - Present*

•Oracle Cloud Infrastructure 2025 Certified Data Science Professional *Valid till 2027*

•Oracle Cloud Infrastructure 2025 Certified Generative AI Professional *Valid till 2027*

•Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate *Valid till 2027*